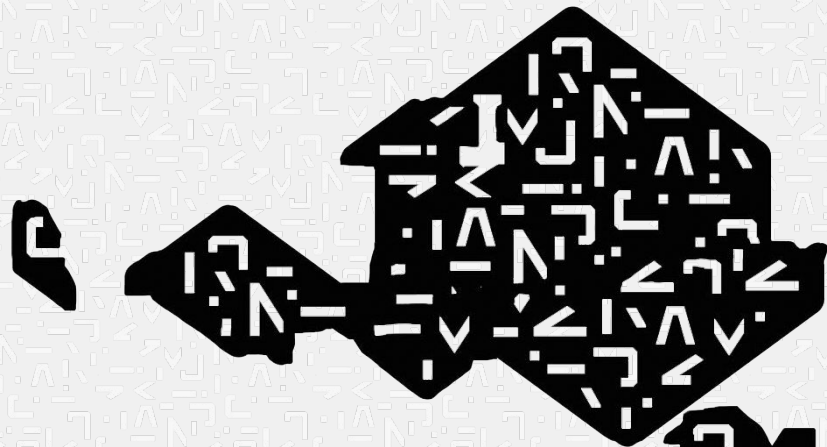
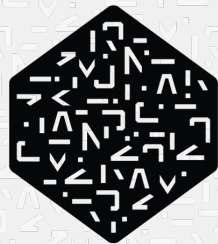


NUN



הַנּוֹתָן לַיָּם
וְלָאֵשׁ וְלָאֵרֶץ
וְלָכָל חַיָּה
וְלָכָל בְּרִיאַת
אֶרֶץ מִצְרָיִם
וְלָכָל בְּרִיאַת
אֶרֶץ מִצְרָיִם

Introduction to Financial Machine Learning



Jon R. Taylor, Ph.D. (Arbitrage)

Yo Arb, what have you been
up to?



Disclaimer

Nothing in this presentation should be considered as financial advice. Staking with NMR is meant to be an indicator of **your confidence** in your machine learning model. Participants may lose NMR staked in tournaments. **Do your own research** before obtaining NMR!

Disclaimer

This presentation reflects my own views and opinions and do not reflect the views or opinions of the Federal Reserve Bank of St. Louis or the Federal Reserve Board of Governors

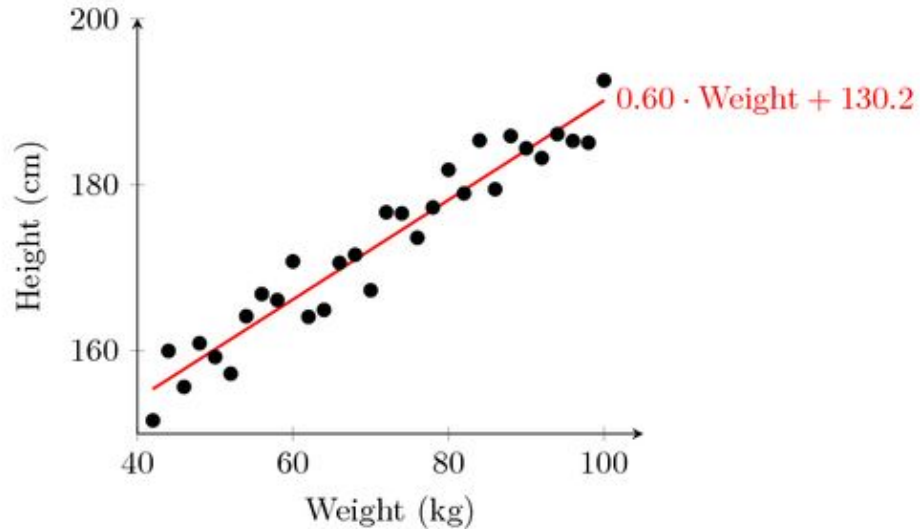
“An approximate answer to the right problem is worth a good deal more than an exact answer to an approximate problem.”

-John Tukey

What is Linear Regression?

- This presentation provides a “Big Picture” introduction of Linear Regression in a Machine Learning setting
- We'll review:
 - What is Linear Regression?
 - How do we evaluate the output of a Regression model?
 - Tips and Tricks
- By the end of this presentation, you'll be familiar with the basics of Linear Regression in a Machine Learning setting

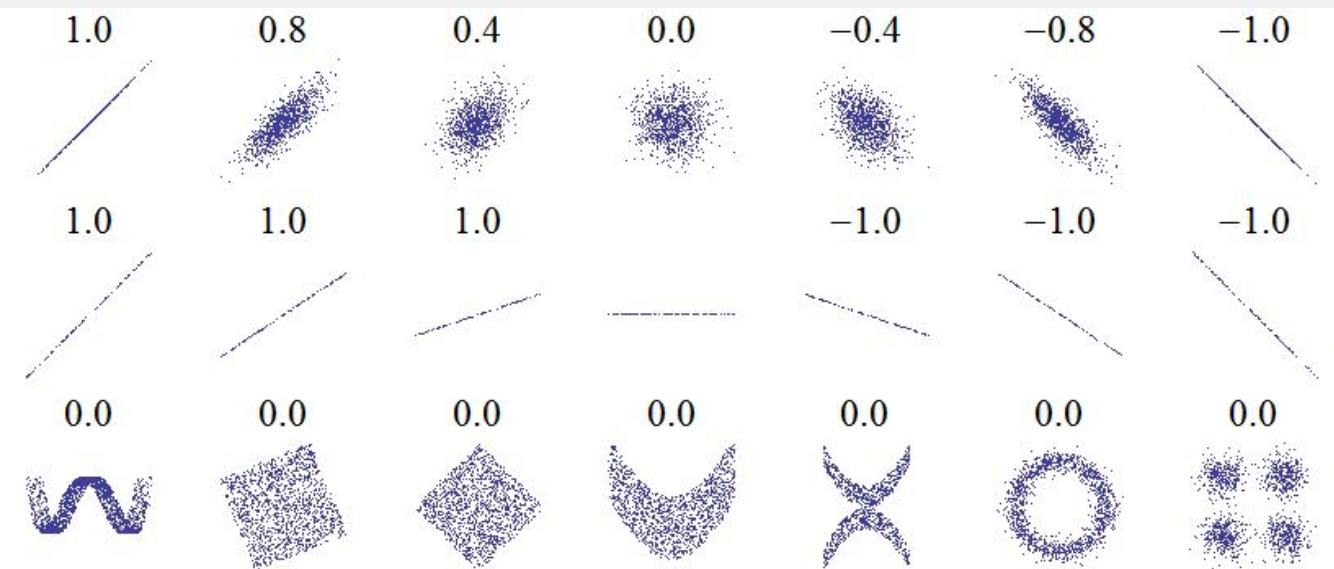
What is Linear Regression?



Is the Y-axis linearly related to the X-axis?

YES – the SLOPE (0.6) times the observed value of X at a given point is approximately equal to the Y value, plus the constant (130.2)

What is Linear Regression?



Correlation matters!

What is Linear Regression?

Table 3 Average Marginal Effects

Average marginal effects (AME) from estimating a logistic regression model to explain bank failures, where the dependent variable FAIL takes on a value of one if a bank failed during 2009, and a value of zero otherwise. Explanatory variables are defined in Table 1. There are 115 banks that were closed by the FDIC during 2009 and 6,510 survivors when we use year-end 2008 data.

CW Model Fit on 2008 Data to Predict 2009 Failures						
Variable	dy/dx	Std. Err.	z	Pr(> z)	Conf. Int.	Cont. Int.
TE	-0.618	0.064	-9.653 ***	0.000	-0.723	-0.513
LLR	-0.093	0.097	-0.952	0.341	-0.253	0.067
ROA	-0.284	0.059	-4.851 ***	0.000	-0.381	-0.188
NPA	0.100	0.019	5.163 ***	0.000	0.068	0.132
SEC	-0.023	0.014	-1.681 *	0.093	-0.046	0.000
BD	-0.003	0.008	-0.399	0.690	-0.017	0.010
LNSIZE	0.001	0.001	2.136 **	0.033	0.000	0.003
CASHDUE	-0.009	0.022	-0.397	0.691	-0.043	0.028
GOODWILL	0.572	0.116	4.918 ***	0.000	0.381	0.763
RER14	-0.043	0.013	-3.231 ***	0.001	-0.065	-0.021
REMUL	0.038	0.020	1.915 *	0.056	0.005	0.071
RECON	0.022	0.015	1.512	0.131	-0.002	0.046
RECOM	0.002	0.011	0.173	0.862	-0.017	0.021
CI	-0.022	0.018	-1.209	0.227	-0.051	0.008
CONS	-0.034	0.044	-0.782	0.434	-0.106	0.038

*, **, *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively

Sometimes, more than one variable is associated with a change in the dependent variable!

Though we may not know which “feature” is statistically significant ahead of time, we can let an algorithm figure it out for us. That is what machine learning does!

How do we evaluate the output of a regression model?

Evaluating models is a science unto itself. Thankfully, the field is very robust and there are many ways to evaluate the performance of a model.

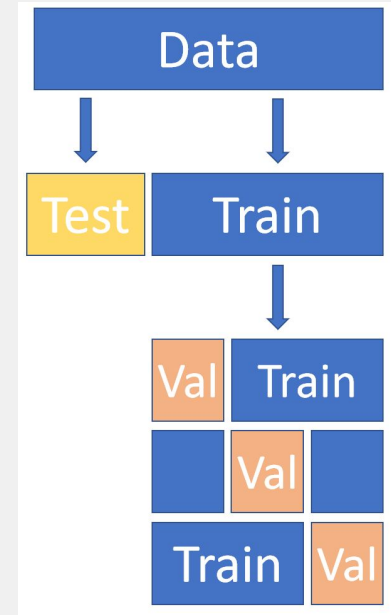
Here are 3 examples for two types of evaluations:

In Sample	Out of Sample
R-squared	Correlation
Akaike Information Criterion	Log-loss
F-Statistic	ROC-AUC

How do we evaluate the output of a regression model?

What does in-sample and out-of-sample mean?

If you have enough data, you can “hold out” some of your data to evaluate how your model performs “out of sample.”

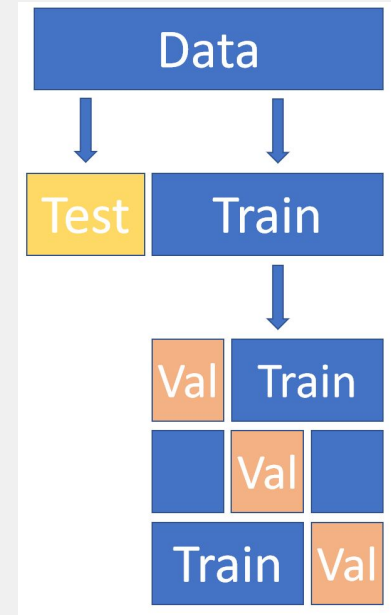


How do we evaluate the output of a regression model?

What does in-sample and out-of-sample mean?

For instance, if you had 100 observations, you could “train” your model on 70 of the observations, and then predict on the remaining 30, and compare predicted values to actual values.

If your model is very good, then the difference is zero!



Tips and Tricks

1. Always explore your dataset before you run any models.
 - a) Look for outliers
 - b) Check for missing values
 - c) Graph the data to look for clues

Tips and Tricks

2. Think about your research design before running any models
 - a) Do you have enough data to perform out of sample tests?
 - b) Are there any obvious groupings to consider?
 - c) Does your data have non-linear relationships?

Tips and Tricks

3. Occam's Razor

Sometimes, the simplest explanation is the correct one!

Not every machine learning problem requires Neural Networks, Genetic Programming, or other “deep learning” techniques.

Tips and Tricks (Remember this?)

Sometimes, more than one variable is associated with a change in the dependent variable!

Table 3 Average Marginal Effects

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Tips and Tricks (Remember this?)

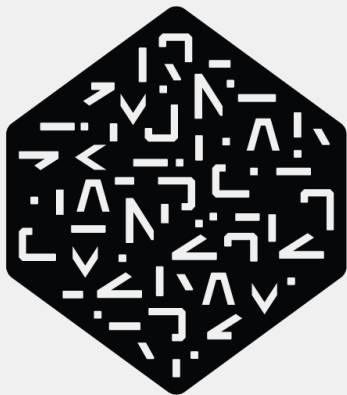
This was not a linear model! Often, non-linear data can be transformed in such a way so that a linear relationship can be examined. Always think “outside the box” BEFORE you run any models.

Table 3 Average Marginal Effects

Average marginal effects (AME) from estimating a logistic regression model to explain bank failures, where the dependent variable FAIL takes on a value of one if a bank failed during 2009, and a value of zero otherwise. Explanatory variables are defined in Table 1. There are 115 banks that were closed by the FDIC during 2009 and 6,510 survivors when we use year-end 2008 data.

CW Model Fit on 2008 Data to Predict 2009 Failures						
Variable	dy/dx	Std. Err.	z	Pr(> z)	Conf. Int. Low	Cont. Int. Hi.
TE	-0.618	0.064	-9.653 ***	0.000	-0.723	-0.513
LLR	-0.093	0.097	-0.952	0.341	-0.253	0.067
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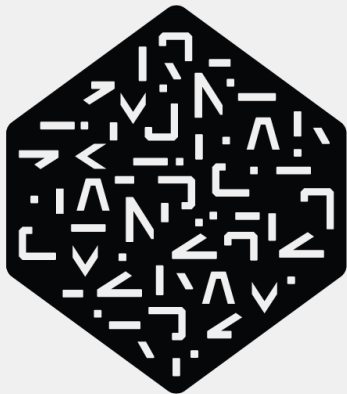
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NUMERAI

The hardest data science tournament
on the planet.

Also the highest paying.

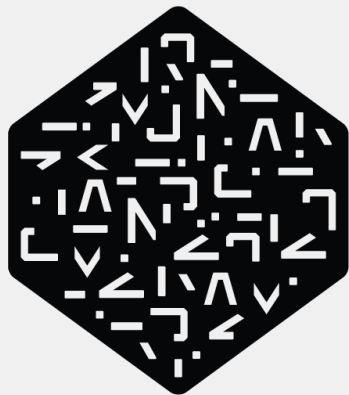


NUMERAI

In general, a balanced approach is better.

Don't overfit

Cross-validate with Eras in mind.



NUMERAI

Have fun!

Evaluating the Performance of Machine Learning Models

Evaluating the Performance of Machine Learning Models

Find the next number of the sequence

1, 3, 5, 7, ?

Evaluating the Performance of Machine Learning Models

Correct solution

217341

because when

$f(x) = \frac{18111}{2}x^4 - 90555x^3 + \frac{633885}{2}x^2 - 452773x + 217331$

VIA 9GAG.COM

Evaluating the Performance of Machine Learning Models

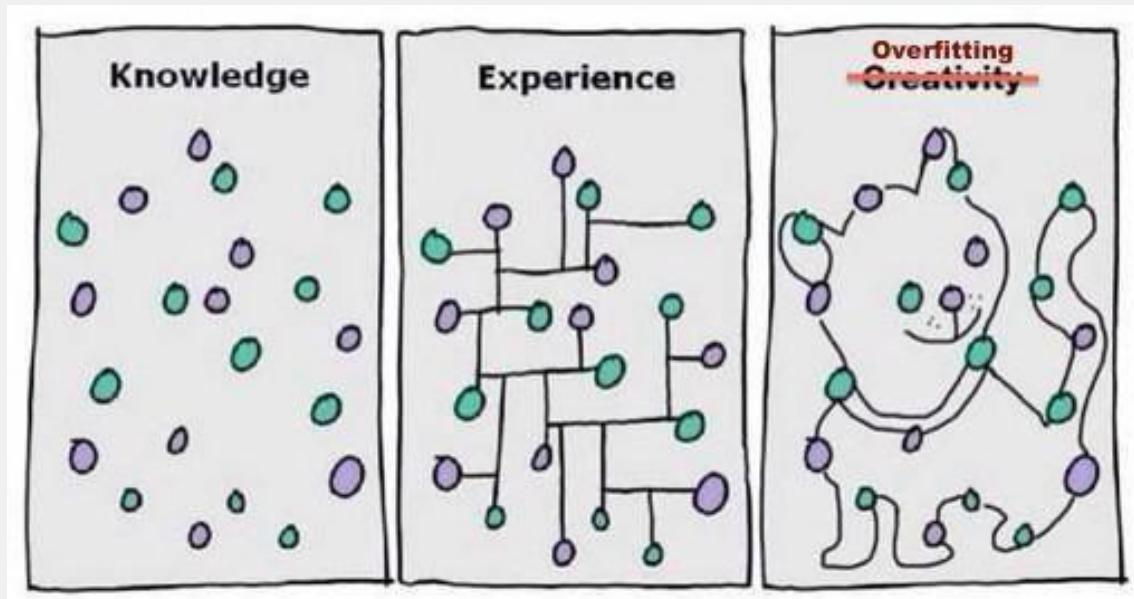
$f(1)=1$
 $f(2)=3$ much solution
 $f(3)=5$ wow very logic
 $f(4)=7$
 $f(5)=217341$
such function
many maths
wow

A Shiba Inu dog, commonly known as Doge, is shown in the bottom right corner of the image. The dog is looking upwards and to the left with a curious expression. The entire image has a bright cyan background.

Evaluating the Performance of Machine Learning Models

- This part of the presentation provides a “Big Picture” view of evaluating the performance of a machine learning model
- We'll review:
 - What overfitting means
 - How to avoid overfitting
 - How to measure the performance of a model
- By the end of this lecture, you'll be familiar with the basics of evaluating the performance of a machine learning model

What is Overfitting?

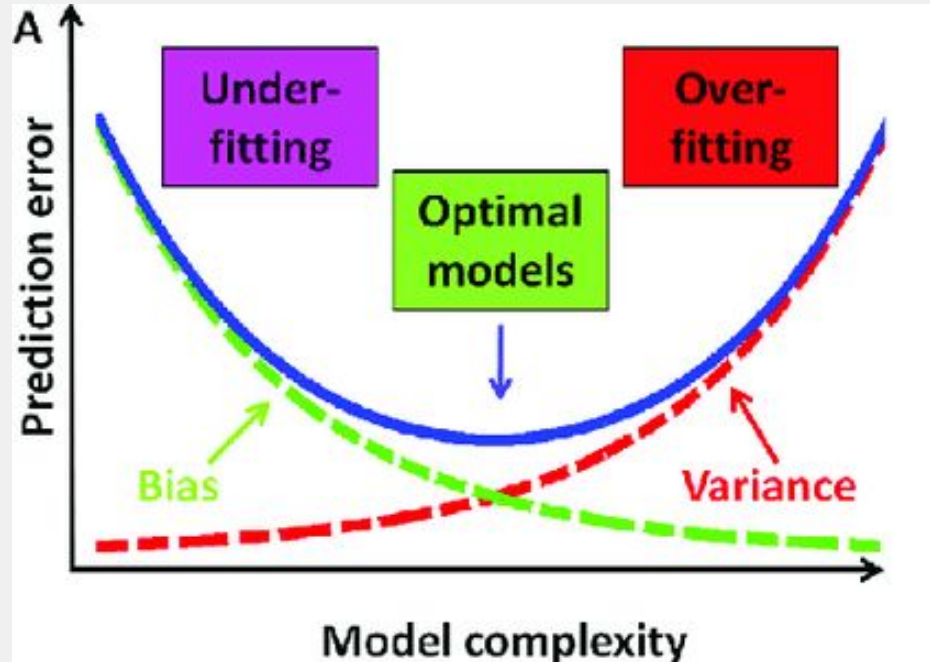


What is Overfitting?

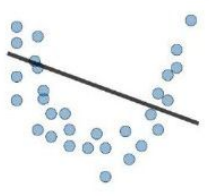
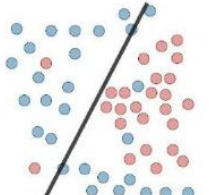
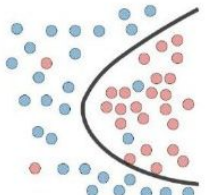
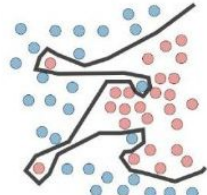
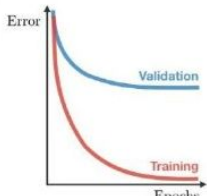
Overfitting is when you have a complicated model that gives worse predictions, on average, than a simpler model.

What is Overfitting?

Overfitting is when you have a complicated model that gives worse predictions, on average, than a simpler model.



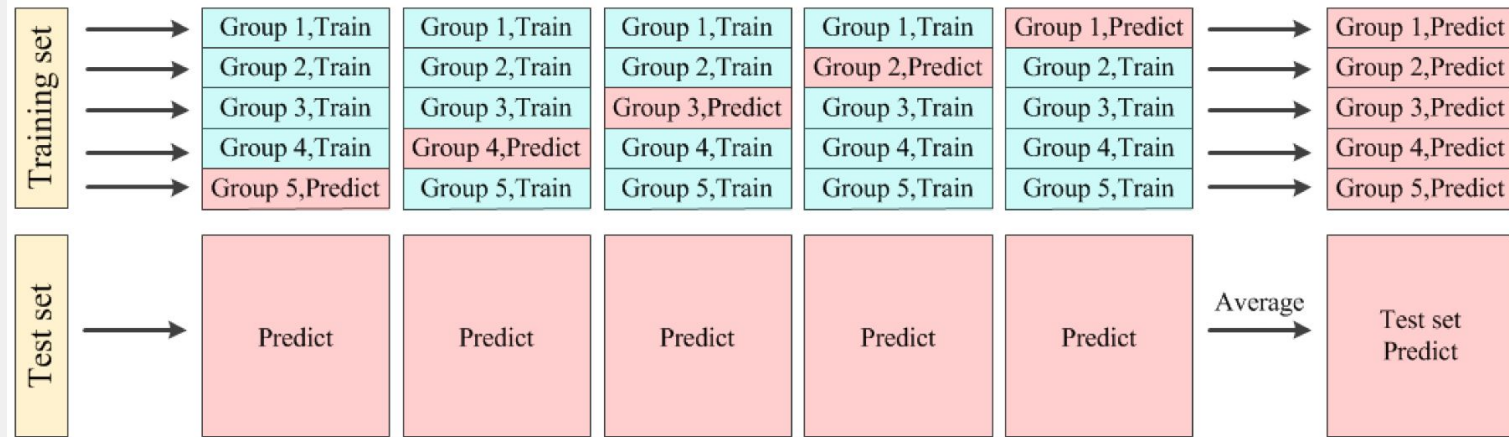
What is Overfitting?

	Underfitting	Just right	Overfitting
Symptoms	- High training error - Training error close to test error - High bias	- Training error slightly lower than test error	- Low training error - Training error much lower than test error - High variance
Regression			
Classification			
Deep learning			
Remedies	- Complexify model - Add more features - Train longer		- Regularize - Get more data

How do we avoid overfitting?

In general, the best way to avoid overfitting is to use cross-validation.

What is cross-validation?



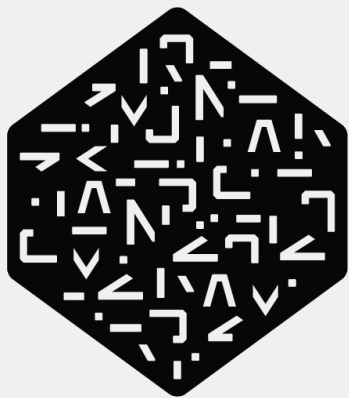
How do we measure performance of a model?

If we have a large enough sample, then we can “hold out” some of the data to evaluate a model’s performance.

There’s a catch!

The data used for evaluation **MUST** be homogenous to the entire sample

If your evaluation data is different from your training data, then you will make bad inferences!



NUMERA I

Do you have any questions?

Now See This!

YOU DON'T HAVE TO CREATE A FULL END-TO-END
MACHINE LEARNING PIPELINE

OCCAM'S RAZOR APPLIES TO FEATURES

TAKE MOMENTUM FOR INSTANCE

Now See This!

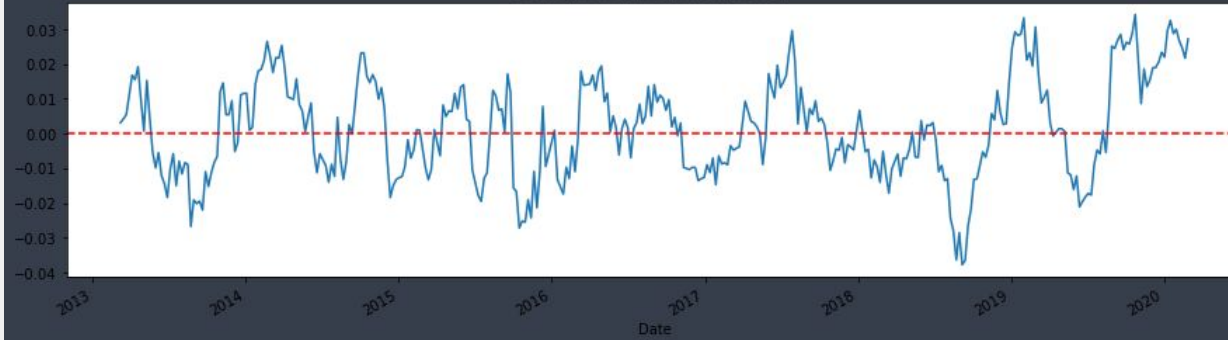
```
eod_merged['6day_ret'] = eod_merged.groupby('ticker')['Adj_Close'].transform(lambda x: (x.shift(6) - x) / x)
eod_merged['ln6day_ret'] = eod_merged.groupby('ticker')['Adj_Close'].transform(lambda x: np.log(x) - np.log(x.shift(6)))
eod_merged_weekly['ln6day_ret'].fillna(0, inplace=True)
eod_merged_weekly['6day_ret'].fillna(0, inplace=True)
eod_merged_weekly['logreturns'] = np.log(1+eod_merged_weekly['ln6day_ret'])
eod_merged_weekly['rolling12'] = eod_merged_weekly.groupby('ticker')['logreturns'].apply(lambda x: x.rolling(24, min_periods=24).sum())
eod_merged_monthly['cumret'] = np.exp(eod_merged_monthly.rolling12)-1
eod_merged_monthly['momentumrank'] = eod_merged_monthly.groupby('Date')['cumret'].rank(method='first', pct=True, ascending=False)
```

Mean Correlation: 0.0015
Median Correlation: -0.0006
Standard Deviation: 0.0487

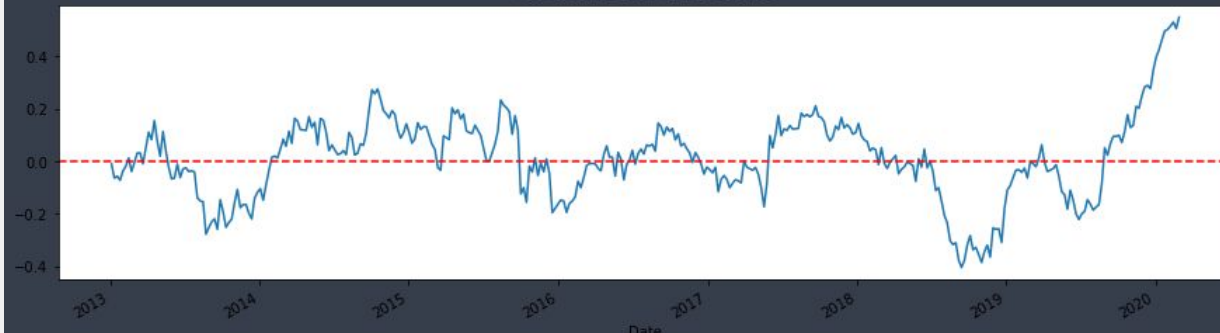
Mean Pseudo-Sharpe: 0.2238
Median Pseudo-Sharpe: -0.0959

Hit Rate (% positive eras): 49.59%

Rolling Per Era Correlation Mean



Cumulative Sum of Era Scores



Now See This!

```
eod_merged_monthly['key'] = eod_merged_monthly.date.astype(str) + eod_merged_monthly['BB Ticker']
full_data = eod_merged_monthly.join(targets.set_index('key'), on='key', how='left', rsuffix='t_', lsuffix='q_')

full_data = targets.join(eod_merged_monthly.set_index('key'), on='key', how='left', lsuffix='_y', rsuffix='_x')

full_data.dropna(subset=['friday_date_y'], inplace=True)

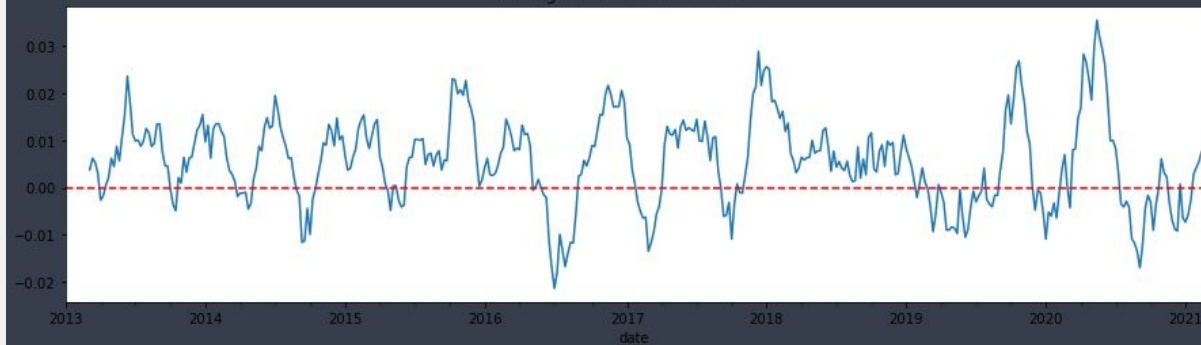
full_data.fillna(0.5, inplace=True)
```

Mean Correlation: 0.0056
Median Correlation: 0.0068
Standard Deviation: 0.0280

Mean Pseudo-Sharpe: 0.1982
Median Pseudo-Sharpe: 0.2416

Hit Rate (% positive eras): 59.91%

Rolling Per Era Correlation Mean



Cumulative Sum of Era Scores

